

Lingam Hemang

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Professional Summary

GenAI Engineer who has **designed and shipped production AI applications** for Apple Inc. over 4 years at Infosys – architecting end-to-end **RAG pipelines, LLM-powered APIs, and agentic AI tools** using **LangChain, AWS Bedrock, and ChromaDB** from first design through deployment. Every AI system I build sits on data infrastructure I also own: distributed pipelines on **Spark, Kafka, Airflow, and Snowflake** that I engineered to be reliable enough to trust for grounding production LLM applications.

Technical Skills

GenAI & LLM: AWS Bedrock, LangChain, RAG Pipelines, Vector Embeddings, ChromaDB, Ollama, Prompt Engineering, LLM Application Development, Agentic AI, SSE Streaming

Languages: Python, SQL, PySpark

Backend & APIs: Django, Flask, REST API Design & Integration, Server-Sent Events (SSE), OOP, SOLID Principles, Design Patterns, pytest

Data Engineering: Apache Spark, Apache Kafka, Apache Airflow, Snowflake, AWS S3, AWS Lambda, ETL Pipelines, Data Modelling, Schema Design & Validation, Partitioning & Query Optimisation

ML: XGBoost, LightGBM, scikit-learn, Plotly, Dash

DevOps & Tools: Docker, Git, CI/CD, Linux, Bash, AWS Lambda, Splunk, Agile/Scrum, Jira, Confluence

Projects

Agentic AI Git Diff Analyzer | *Python, Django, AWS Bedrock, AWS Lambda, Ollama, ReportLab* | [Repo](#)

- Architected a **dual-mode agentic SQL schema impact analyzer** – local (Ollama) and cloud mode (**AWS Bedrock Claude 3.5** via GitHub Webhook → Lambda → SSM → EC2); designed commit-accurate downstream dependency detection and **LLM-generated impact reports** with risk-level badges (CRITICAL → LOW) as styled A4 PDFs; built real-time **Django UI** with **SSE** streaming and run deduplication by commit hash.

Document Intelligence Chatbot – RAG Pipeline | *Python, Django, LangChain, ChromaDB, Ollama, EasyOCR* | [Repo](#)

- Designed an **end-to-end RAG pipeline** (LangChain + ChromaDB + Ollama) with persistent multi-turn conversation memory; built multi-format document ingestion (PDF, DOCX, TXT, Images) via **PyMuPDF + EasyOCR** with auto-rotation; architected an incremental vector store with timestamped snapshots/rollback and a **10-endpoint Django REST API** supporting live model switching and VectorDB management.

Retail ML Sales Forecasting API | *Python, Flask, scikit-learn, XGBoost, LightGBM* | [Repo](#) **Infosys Hackathon – 2nd Place**

- Built a **multi-model ML forecasting system** (Random Forest, XGBoost, LightGBM) for sales, store projections, and product recommendations via **Flask REST API**; designed cyclical sin/cos month encoding for seasonal patterns and cross-join **purchase probability scoring**; shipped an interactive **Dash + Plotly** dashboard.

Experience

Infosys

Hyderabad, India

Specialist Programmer – GenAI Engineer

July 2022 – Present

Client: *Apple Inc.* | **Python, Spark, Kafka, Airflow, Snowflake, AWS S3, AWS Lambda, PySpark, Splunk**

- Architected production **ETL pipelines** on a large-scale distributed platform (**Spark, Kafka, Airflow, AWS**) – the reliable data foundation my AI systems are built on; designed Python data services consuming **REST APIs** into **Snowflake**, cutting processing time **40%**.
- Designed **fault-tolerant ingestion pipelines** across **AWS S3** and **Kafka** via **Airflow DAGs**; engineered retry logic and self-healing schema validation from scratch – sustaining **99.5% uptime** and cutting error rate **35%**.
- Independently productionised a client **ML name-matching model** onto the Airflow cluster – owned input-view provisioning, connection configuration, and execution management; architected **semantic layer** materialised views feeding downstream analytics and Tableau dashboards.
- Owned **data consistency** for a production application – designed database view validation logic and the load process into a **semantic search cluster**; identified an onboarding bottleneck and built **reusable framework extensions**, cutting onboarding effort **30%**.

Education

SRM Institute of Science and Technology

May 2018 – June 2022

Bachelor of Technology – Electronics and Communication Engineering

CGPA: 9.52/10.0